CNC MACHINE OPERATIONS & CARE

CNC Preventive Maintenance Manual

Daily • weekly • monthly • scheduled maintenance concepts

DOCUMENT CONTROL	DETAIL
Document	CNC_Preventive_Maintenance_Manual.pdf
Edition	1.0 — General reference edition
Prepared	11 November 2026
Applicability	General CNC machining centres, lathes and routers; adapt locally

Important: This is an educational, generic reference. CNC machines differ substantially by builder, model, control, options, tooling and site rules. The machine-specific operator, maintenance, electrical and safety manuals take precedence. Do not use example values, procedures or alarm descriptions as substitutes for approved machine documentation.

1. Purpose and maintenance philosophy

Preventive maintenance (PM) is planned inspection, cleaning, lubrication, adjustment and condition monitoring intended to reduce unplanned downtime and preserve machine accuracy. This schedule is a starting framework only. Actual intervals, lubricants, quantities, torque values, filter grades and service procedures must come from the machine builder and site maintenance plan.

Safety before maintenance

Apply the site lockout/tagout (LOTO) procedure before servicing, removing guards, clearing jams or entering hazardous areas. Identify and control electrical, pneumatic, hydraulic, gravity, spring, thermal and rotational stored energy. Verify isolation and zero-energy state using the approved method. Only qualified personnel perform electrical and energy-system work.

2. Daily / per-shift checks

Item	Typical task	Record / escalation
General condition	Check for leaks, abnormal noise, vibration, smell, loose covers, damaged cables or visible damage.	Report abnormality; do not operate with a safety-critical defect.
Safety devices	Check guards, door operation, E-stops and interlocks using the operator checklist; never bypass.	Record pass/fail. Remove from service if a safety function fails.
Lubrication	Check automatic lubrication reservoir/level and low-level indicators; verify no active lubrication alarm.	Top up only with specified lubricant and approved method.
Coolant	Check level, visible contamination, concentration using the site method, flow and nozzle direction.	Correct per coolant supplier/site procedure; use SDS and PPE.
Chip management	Remove chips from designated accessible areas; inspect chip conveyor and bin capacity.	Stop and isolate before clearing jams or reaching into danger zones.
Air supply	Check indicated pressure, water separator/drain status and visible leaks.	Do not adjust pressure outside OEM range.
Work area	Check floor, access, lighting, housekeeping and spill condition.	Keep exits and electrical panels unobstructed.

3. Weekly checks

- Clean accessible enclosure surfaces, windows, door tracks and approved way covers; avoid directing high-pressure air at seals, bearings, sensors or electrical components.
- Inspect coolant concentration, pH/condition and tramp-oil condition according to the coolant supplier and site limits.
- Inspect accessible hoses, fittings, cable carriers and protective covers for wear, abrasion, cracks or leakage.
- Check toolholders, pull studs, collets, chuck jaws and fixture contact surfaces for damage and contamination.
- Inspect chip conveyor operation and clean the accessible filter/screen according to the isolation procedure.
- Review alarms, lubrication consumption, coolant use and downtime trends; open work orders for recurring issues.

4. Monthly checks

- Inspect and clean specified filters, strainers and coolant tank areas using the approved procedure.
- Check way-lube, hydraulic and pneumatic system indicators; investigate unusual consumption or pressure fluctuation.
- Inspect spindle taper, tool interface and drawbar/clamping indicators using OEM-approved checks.
- Check enclosure seals, door rollers, wipers and way covers for wear or damage.

- Review axis and spindle load trends, temperature trends and repeatability measurements where a baseline exists.
- Verify that maintenance records, calibration status and spare-part stock are current.

5. Scheduled / periodic maintenance

Frequency (example only)	Typical work scope	Who
Quarterly / 500 operating hours	Inspect lubrication distribution, coolant system condition, chip conveyor, air preparation and visible electrical cabinet condition (external only unless qualified). Review recurring alarms.	Maintenance technician
Semi-annually / 1,000 hours	Perform OEM-defined geometry/repeatability checks; inspect belts, couplings, filters, fans and cooling systems; verify clamping and tool-change function.	Qualified maintenance / service
Annually / OEM interval	Complete comprehensive inspection per builder schedule, including accuracy checks, safety validation, fluid replacement where specified, backup verification and condition assessment.	Authorized service / competent team
After collision / overload / major repair	Hold machine from production pending documented inspection of affected axes, spindle, tooling, workholding, guards and safety functions.	Authorized service / engineering

Do not convert the example time periods into mandatory intervals. Use whichever interval the OEM specifies, and adjust only through an approved risk- and condition-based maintenance process.

6. Fluid, lubricant and consumable control

- Use only the exact lubricant, hydraulic fluid, coolant and grease grades approved for the machine and application.
- Do not mix products unless the supplier and machine builder explicitly confirm compatibility.
- Label containers, keep them clean and sealed, and prevent cross-contamination.
- Follow safety data sheets (SDS), exposure controls, disposal rules and local environmental requirements.
- Record product name, batch where required, quantity, date and person performing the service.

7. Maintenance work controls

1. Review the work order, OEM procedure, drawings and risk assessment before starting.
2. Notify affected operators and remove the machine from production; secure tools, workpiece and movable components.
3. Apply LOTO to all relevant energy sources. Release or restrain stored energy and verify zero energy.
4. Use correct tools, lifting equipment and PPE. Protect precision surfaces and keep contamination out of open systems.
5. Complete the task, restore guards and connections, account for tools and materials, and remove LOTO only under site rules.
6. Test safety functions and machine operation using the authorized post-maintenance test procedure.
7. Document parts, measurements, adjustments, outstanding defects and formal release to production.

8. Maintenance log template

Date / machine hours	Task / checklist item	Reading / finding	Action / parts	Technician / verification
_____	_____	_____	_____	_____
_____	_____	_____	_____	_____

Date / machine hours	Task / checklist item	Reading / finding	Action / parts	Technician / verification
_____	_____	_____	_____	_____
_____	_____	_____	_____	_____
_____	_____	_____	_____	_____

9. Suggested PM performance measures

- PM completion rate: completed planned tasks ÷ tasks due.
- Repeat-fault rate: faults recurring within a defined period after repair.
- Unplanned downtime and mean time to repair (MTTR).
- Lubricant/coolant consumption anomalies and leakage trends.
- Accuracy or repeatability drift against the machine's accepted baseline.

10. Release criteria

A machine is ready for production only when the maintenance task is complete, all guards and safety devices are restored, required functional checks pass, defects are closed or formally risk-assessed, and an authorized person records release.